

Unit – 1 : Session – 7 : SLO – 2 | Class Design & Inheritance**Activity (Solutions)**

- 1) `class Vehicle { String brand; void move(){ System.out.println("moving"); } }` `class Car extends Vehicle { int doors; }` `class ElectricCar extends Car { int batteryKwh; void charge(){ System.out.println("charging"); } }` — multilevel hierarchy Vehicle → Car → ElectricCar.
- 2) `class Employee { String name; double salary; void work(){ System.out.println("working"); } }` `class Manager extends Employee { String department; void conductMeeting(){ System.out.println("meeting"); } }` `Manager m = new Manager(); m.work(); m.conductMeeting();`
- 3) Overriding + super: `class Employee { void show(){ System.out.println("Employee"); } }` `class Manager extends Employee { @Override void show(){ super.show(); System.out.println("Manager"); } }` `new Manager().show();` prints Employee then Manager — `super.show()` invokes the superclass version, letting the subclass extend rather than fully replace the parent behavior.